

2.5 恐龙的追逐捕食模型(英文)

问题

velociraptor 是生活在距今约 7500 万年前后白垩纪(译注:白垩纪为距今 1.36~0.65 亿年的地质年代,是中生代最后的纪)的一种食肉(捕食其他动物的)恐龙。

古生物学家认为这是一种非常顽强的猎食其他动物的野兽,而且可能是成对或成群地外出追猎。然而,不幸的是无法像观察现代哺乳食肉动物在野外是如何追猎其食物的行为那样观察到 **velociraptor** 在野外的追猎行为。一组古生物学家来到你们队请求你们在 **velociraptor** 的追猎行为的建模方面给予帮助,他们希望把你们的结果与研究狮子、老虎及其他类似的食肉动物行为的生物学家的研究报告相比较。

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平均的成年 *velociraptor* 长 3m, 髌高 0.5m, 重约 45kg. 据估计, 这种动物跑得非常快, 速度可达 60km/h, 持续约 15s. 在以这种速度开始的冲刺后, 它要停下来在其肌肉中增加乳酸以恢复体力。

假设 *velociraptor* 捕食一种称为 *thescelosaurus* (太西龙属) 的大小与 *velociraptor* 差不多的双足食草动物。从 *thescelosaurus* 化石的生物力学分析得知 *thescelosaurus* 可以 50km/h 的速度长时间奔跑。

第一部分

假设 *velociraptor* 是一只独居的猎食其他动物的野兽, 试设计一个单个的 *velociraptor* 潜近猎物并追猎一只单个的 *thescelosaurus* 的策略以及被追捕物逃避追捕的策略的数学模型。假设当 *velociraptor* 潜近 15m 内时, *thescelosaurus* 总能觉察到, 根据栖息地及气候条件的不同, 甚至在 (多达 50m 的) 更大的范围内觉察欲捕食它的动物的存在。此外, 由 *velociraptor* 的身体结构及体能, 它在全速奔跑时的拐弯半径是有限的, 据估计, 拐弯半径大约是其髌高的 3 倍。另一方面, *thescelosaurus* 却是极其灵活的, 其拐弯半径只有 0.5m。

第二部分

更现实地假设 *velociraptor* 是成对外出追猎, 试设计一个新的关于成对的 *velociraptor* 潜近猎物并追猎一只单个的 *thescelosaurus* 的策略以及被追捕物逃避追捕的策略的数学模型。利用第一部分给出的假定和限制。

Abstract

In this paper, we try to describe the strategies for the *velociraptor* to hunt the *thescelosaurus* and for the *thescelosaurus* to escape. We first establish the basic motion equations.

To Problem I, we develop two models. When response time is neglected, Model I is based on the assumption that the velociraptor always chases in the direction toward the thescelosaurus. Under this condition, the optimal evasive strategy of the thescelosaurus is to run away in the opposite direction the thescelosaurus discovers the velociraptor. In Model II, we let the response time be 0.2 second. We give such evasive strategy that, at the beginning, the thescelosaurus runs in the opposite direction the thescelosaurus discovers the velociraptor; and that, whenever the distance between the velociraptor and the thescelosaurus decreases to l_1 , it turns a certain angle λ and runs in a straight line again. We use the computer to simulate their strategies. We conclude that the thescelosaurus can escape from the velociraptor when $l_1 = 3.2\text{m}$, $\lambda \in [100^\circ, 130^\circ]$.

To Problem II, we conclude that the thescelosaurus will be captured when it discovers the velociraptor within 50 meters, even using the optimal strategy to run in the midperpendicular of the two velociraptors.

Besides, we analyze the stability of the models, and find that the result is considerably influenced by the speed of the thescelosaurus and the response time.

Restatement

The velociraptor was a predatory dinosaur, and it preyed on the thescelosaurus, a herbivorous biped.

The problems needed considered are:

Problem I: Assuming the velociraptor is a solitary hunter, design a mathematical model that describes a hunting strategy

for the velociraptor as well as an evasive strategy of the prey.

Problem I : How about are the conditions of two velociraptors stalking and chasing a single thescelosaurus?

Assuming the thescelosaurus was the same size as the velociraptor. Both were 3 meters long and an approximate mass of 45kg. The velociraptor has the advantage on speed. It could run at a speed of 60 km/h, while 50 km/h for the thescelosaurus. On the other hand, the thescelosaurus is more agile and endurable. Its minimum turning radius is 0.5 meters while 1.5 meters for the velociraptor. And it can run at full speed for long periods of time while 15 seconds for the velociraptor only.

Other conditions are; the thescelosaurus can always detect the velociraptor when it comes within 15 meters, but may detect it at even greater ranges (up to 50 meters) depending upon the habitat and weather conditions.

Basic Assumptions

(1) The area where preying process takes place is broad, smooth and without any obstacles.

(2) The animals are considered as motion particles.

(3) The time for acceleration is neglected.

(4) The animals always keep full speed during the process.

(5) If the thescelosaurus can't be captured in 15 seconds, it will be considered having escaped.

(6) The response time is defined as the interval between the velociraptor's turn and the thescelosaurus's turn.

(7) Once the thescelosaurus detects the velociraptor, it begins to escape, and at the same time the velociraptor begins to

chase.

Symbols

<i>Velo</i>	velociraptor
<i>Thes</i>	thescelosaurus
v_1, v_2	the speed of <i>Velo</i> and <i>Thes</i>
r_1, r_2	the minimum turning radius of <i>Velo</i> and <i>Thes</i>
θ	the intersection angle between the directions of v_1 and v_2
$R_1(t), R_2(t)$	the turning radius function of <i>Velo</i> and <i>Thes</i>
d	the distance between <i>Velo</i> and <i>Thes</i>
t	the response time

Analysis of the Problem

According to the assumption (2), naturally, we expect to establish the equations to describe the relative motion of *Velo* to *Thes*, and find the condition for *Thes* to be captured.

Establishing the Equations of Relative Motion

As indicated in Fig. 2.5.1, we fix the location of *Velo* as the origin of coordinates, and the direction of

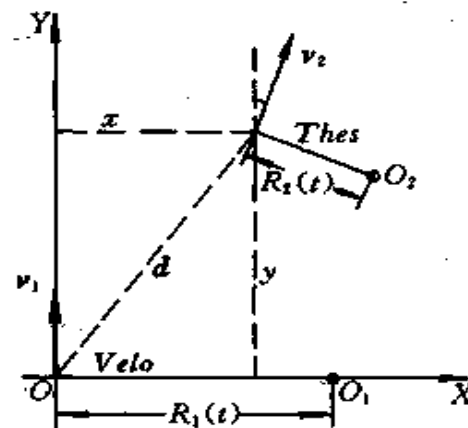


Fig. 2.5.1

Velo's velocity as the direction of the *Y*-axis to establish a moving coordinate system. And O_1, O_2 ; the turning centre of *Velo* and *Thes*. $R_1(t), R_2(t)$; the actual turning radius of *Velo* and *Thes*. When turning clockwise, we have $R_1(t) \geq r_1, R_2(t) \geq r_2$. While anticlockwise, $R_1(t) \leq -r_1, R_2(t) \leq -r_2$.

According to the kinematics, the relative velocity of *Thes* and *Velo* is:

$$v_{\text{relative-velocity}} = v_2 - v_{\text{carrier-velocity}}$$

$v_{\text{carrier-velocity}}$ is the velocity of the spot which coincide with *Thes* in the moving coordinates. It equals to vector sum of the rotating velocity of *Velo* and v_1 . The angular velocity is $v_1/R_1(t)$, hence

$$v_{\text{carrier-velocity}} = v_1 \cdot d/R_1(t) + v_1$$

According to the theorem of projection of velocities, we have

$$\dot{x} = -\frac{v_1}{R_1(t)}y + v_2 \sin \theta$$

$$\dot{y} = \frac{v_1}{R_1(t)}x + v_2 \cos \theta - v_1$$

And the relative angular velocity of *Thes* and *Velo* is $\dot{\theta} = \omega_2 - \omega_1$ where

$$\omega_2 = \frac{v_2}{R_2(t)}, \quad \omega_1 = \frac{v_1}{R_1(t)}$$

hence we have $\dot{\theta} = \frac{v_2}{R_2(t)} - \frac{v_1}{R_1(t)}$

So the equations of relative motion is

$$\dot{x} = v_2 \sin \theta - \frac{v_1}{R_1(t)}y \quad (1)$$

$$\dot{y} = v_2 \cos \theta + \frac{v_1}{R_1(t)}x - v_1 \quad (2)$$

$$\dot{\theta} = \frac{v_2}{R_2(t)} - \frac{v_1}{R_1(t)} \quad (3)$$

The Condition for *Thes* to be Captured

We will define an captured area where we remark *Thes* to be captured when it falls into.

In reality, only when *Thes* is at a short distance from *Velo*, it can be captured. And when the distance is short, it is difficult for *Velo* to turn a large angle to capture *Thes*, due to its high speed and huge structure. So we define the captured area as a sector in front of *Velo*. We show it in Fig. 2. 5. 2.

The coordinate system is the same as that of Fig. 2. 5. 1. The mathematical expression is:

$$x^2 + y^2 - l^2 \leq 0 \quad (4)$$

$$x - ytga \leq 0 \quad (5)$$

$$-(x + ytga) \leq 0 \quad (6)$$

l and α are constants.

We let $l = 1.5m$, $\alpha = \pi/6$, according to their figures and the habits of some animals, e. g. , some beasts like to pounce on the head or the neck of the prey.

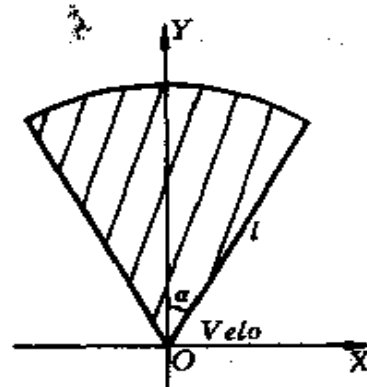


Fig. 2. 5. 2

Development of the Models

To Problem I , we develop Models I and II according to the different response time. Then we discuss the Problem I on the base of Model I .

Problem I

1. Model I : Neglect the response time($t^* = 0$)

In fact, to try its best to chase toward *Thes* is the instinctive

behavior of *Velo*. So we assume that *Velo* always chases towards *Thes* during the whole hunting process.

According to this assumption, *Thes* will always lie in front of *Velo*. So we have $x \equiv 0$

By (1), we have

$$\dot{x} = v_2 \sin \theta - \frac{v_1}{R_1(t)} y = 0$$

hence

$$R_1(t) = \frac{v_1 y}{v_2 \sin \theta}$$

When *Thes* is out of the captured area, by (4), we have $y \geq l$
hence

$$|R_1(t)| = \left| \frac{v_1 y}{v_2 \sin \theta} \right| \geq \frac{v_1 l}{v_2} = 1.8 > r_1$$

It proves that the turning radius of *Velo* can always be met under the assumption. It further shows that the assumption is reasonable when the response time is neglected.

Now we can simplify the equations as

$$\dot{x} = x = 0$$

$$\dot{y} = v_2 \cos \theta - v_1$$

$$\dot{\theta} = \frac{v_2}{R_2(t)} - \frac{v_2 \sin \theta}{y}$$

So we have

$$|v_{\text{relative-velocity}}| = \sqrt{\dot{x}^2 + \dot{y}^2} = |v_1 - v_2 \cos \theta| \geq v_1 - v_2 \quad (7)$$

while $\theta = 0$,

$$|v_{\text{relative-velocity}}| = v_1 - v_2$$

Define d_0 as the original distance between *Velo* and *Thes*, and t' as the time for *Velo* to capture *Thes*. We have

$$d_0 - l = \int_0^{t'} |v_{\text{relative-velocity}}| dt = \int_0^{t'} (v_1 - v_2 \cos \theta) dt$$

$$\geq \int_0^{t'} (v_1 - v_2) dt = (v_1 - v_2) t'$$

hence

$$t' \leq \frac{d_0 - l}{v_1 - v_2} \quad (8)$$

while

$$\theta = 0 \quad t' = \frac{d_0 - l}{v_1 - v_2}$$

It proves that *Velo* can always capture *Thes* in the time up to $\frac{d_0 - l}{v_1 - v_2}$.

And for *Thes*, the optimal evasive strategy is to minimize the $|v_{\text{relative-velocity}}|$. By (7), it is to run away straight in the opposite direction *Velo* is discovered.

According to this, if *Thes* can detect *Velo* when the distance between *Velo* and *Thes* is more than 43.2 meters, it will escape, otherwise will be captured.

2. Model 1 : Consider the response time ($t^* \neq 0$)

From the analysis above, we discover that if $t^* = 0$, *Thes* can't escape by taking the advantage of $r_2 < r_1$. And if the distance *Thes* detects *Velo* is less than 43.2m, *Thes* has no chance to escape. Let $t^* = 0.2s$. Consider such a situation that, at the beginning, *Thes* runs straight in the opposite direction *Thes* discovers *Velo*; and that, when the distance between *Thes* and *Velo* decreases to a certain numerical value (let it be l_1), *Thes* begins to turn with the radius r_2 . At this time, *Velo* can't change the direction of his velocity at once but keep it for a period of response time, then turns until *Thes* is in its direction again. Here the turn-

ing radius of *Velo* is r_1 .

Let β be the intersection angle between v_1 and v_2 at the beginning of *Thes*'s turning. Let λ be the angle that *Thes* turns.

The process of turning above can be resolved into three stages, as indicated in Fig. 2. 5. 3.

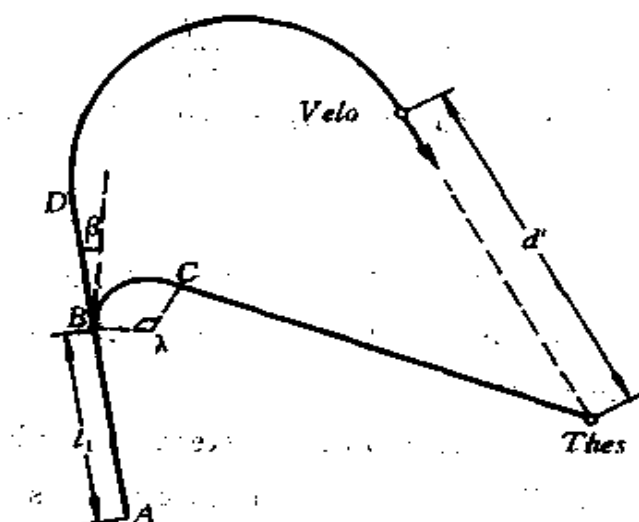


Fig. 2. 5. 3

Stage 1: *Thes* is turning, while *Velo* not. At the beginning of this stage, *Velo* is at *A* and *Thes* is at *B* as showed in Fig. 2. 5. 3. The equations of relative motion are:

$$\dot{x} = v_2 \sin \theta$$

$$\dot{y} = v_2 \cos \theta - v_1$$

$$\dot{\theta} = \frac{v_2}{r_2}$$

The stage will last for $\frac{\lambda}{v_2} r_2$ seconds.

Stage 2: *Thes* finishes turning and runs in a straight line and *Velo* hasn't responded yet. The beginning location of *Thes* is *C*.

The motion equations are:

$$\dot{x} = v_2 \sin \theta$$

$$\dot{y} = v_2 \cos \theta - v_1$$

$$\dot{\theta} = 0$$

The stage lasts for $t^* = \frac{\lambda}{v_2} r_2$ seconds.

Stage 3: *Velo* turns until *Thes* is in its direction again. The beginning location of *Velo* is *D*. The motion equations are:

$$\dot{x} = v_2 \sin \theta - \frac{v_2}{r_1} y$$

$$\dot{y} = v_2 \cos \theta + \frac{v_1}{r_1} x - v_1$$

$$\dot{\theta} = -\frac{v_1}{r_1}$$

Suppose λ , β and l_1 are given, run a C language program on computer using Euler's polygonal arc method, we will compute the distance d' between *Thes* and *Velo* after above process has completed. If $d' > l_1$, it shows that *Thes* can increase the distance from *Velo* by turning and will escape finally.

The results are showed in Tab. 2. 5. 1. We give out the range of λ when β , l_1 are specified. For example, let $\beta = 0^\circ$, $l_1 = 3.0\text{m}$, we have $d' > l_1$, when $\lambda \in [90^\circ, 130^\circ]$.

According to Tab. 2. 5. 1, we give *Thes* such an evasive strategy that to turn an angle ranging from 100° to 130° , whenever the distance between *Thes* and *Velo* is 3.2 meters.

Tab. 2. 5. 1

β $\lambda(^{\circ})$ l_1 (m)	0°	10°	20°	30°	40°	50°	60°	70°	80°	90°
3. 0	90~	80~	60~	50~	40~	30~	20~	10~	0~	0~
	130	120	110	90	80	70	60	40	30	20
3. 2	100~	90~	70~	60~	50~	40~	30~	20~	0~	0~
	130	120	110	100	90	70	60	50	40	30
3. 4	120~	100~	80~	70~	60~	50~	40~	20~	10~	0~
	140	130	120	100	90	80	70	50	40	30
3. 6	130~	110~	100~	80~	70~	60~	50~	30~	20~	10~
	140	130	120	110	100	80	70	60	50	30
3. 8	~	130	110~	100~	80~	70~	60~	50~	40~	20~
			120	110	100	90	70	60	50	40
4. 0	~	140	130	110	100	80~	70~	60	50	40
						90	80			

Problem 1

Consider that two velociraptors hunt a single thescelosaurus, we neglect the response time and assume that the two velociraptors *Velo1* and *Velo2* always sponsor chasing at the same time when *Thes* appears in the middle of them.

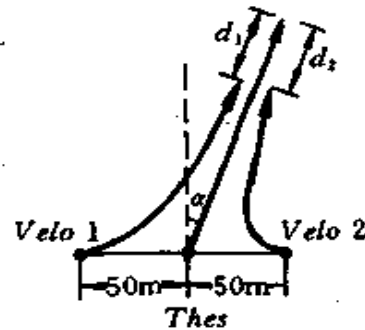


Fig. 2. 5. 4

Assume that *Thes* runs in a straight line and changes its direction while $\min(d_1, d_2) \leq d_0$. Let α be the intersection angle between *Thes*'s direction and the midperpendicular of *Velo1* and *Velo2*, as indicated in Fig. 2. 5. 4.

In order to acquire the optimal angle of α , let the original distance between *Velo1* and *Thes* be 50 meters, the same as that between *Velo2* and *Thes*. Using Euler's polygonal arc method, we compute the time for *Velo* to capture *Thes* in Tab. 2. 5. 2.

Tab. 2. 5. 2

α	0°	10°	20°	30°	40°
Capture time (s)	9.28	7.86	6.48	5.19	4.02

So the optimal angle of α for *Thes* is 0°.

If $\min(d_1, d_2) \leq d_0$, *Thes* will be captured in $\frac{d_0 - l}{v_1 - v_2}$ seconds.

Assume that $\min(d_1, d_2) = d_0$, then, if $\frac{d_0 - l}{v_1 - v_2} < 15 - t_0$ i. e. $t_0 <$

$15 - \frac{d_0 - l}{v_1 - v_2}$, *Thes* will be captured. Here t_0 means the time from the beginning of hunting to when $\min(d_1, d_2) = d_0$.

So we compute the value of t_0 , the results are as the follow Tab. 2. 5. 3.

Tab. 2. 5. 3

d_0 (m)	5	10	20
t_0 (s)	8.02	6.22	3.00
$15 - \frac{d_0 - l}{v_1 - v_2}$	13.74	11.94	8.34

It is to say, even if *Thes* changes its direction much earlier, it cannot escape. So, in the case of hunting in pairs, even if *Thes* can discover the velociraptors within 50 meters, it has no chance to escape.

Stability Analysis

We test Model I in Problem 1 by changing v_1 , v_2 , r_1 , r_2 , t^* . The changing rate of d' is shown in Tab. 2.5.4, when v_1 , v_2 , r_1 , r_2 , t^* are changed respectively. Our conclusion is that v_2 , t^* have greater influence than v_1 , r_1 , r_2 . It means that changing v_2 , t^* has more effects on the strategies of the velociraptor and the thescelosaurus.

Tab. 2.5.4

Changing rate	v_1	v_2	r_1	r_2	t^*
-10%	+5.34%	-16.28%	-2.47%	+1.37%	-11.51%
-5%	+2.42%	-8.10%	-1.24%	+0.68%	-5.82%
+5%	-1.95%	+8.03%	+1.26%	-0.68%	+5.93%
+10%	-3.46%	+16.0%	+2.52%	-1.35%	+11.95%

Strengths and Weaknesses

The greatest strength of our models is its flexibility. Not only can we apply them to studying the hunting behaviors of lions, tigers and similar predatory animals; but also to other similar practical problems, such as collision avoidance.

Another important strength is that we fully discussed the case of solitary hunting, and the results we acquired are quite accurate.

The models are also valuable because they are not quite complicated. In fact, the process of deducing the results in Model I is so easy that it can be understood without any profound

knowledge.

Unfortunately, our models have a few minor weaknesses as well. Being short of some realistic materials, the form of our models rely more on some assumptions than on others. And due to its complexity, we have only made a qualitative analysis on Problem I.

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